

Single-Nomination Impartial Selection: Exact Finite LP Certificates and a Sharp Analysis of the Two-Path Relaxation

A rigorous correction and computational note

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Abstract

Status: partial progress, not a solution of the asymptotic problem. We revisit the open question whether randomized impartial selection in the single-nomination model admits approximation guarantees converging to $3/4$. The asymptotic question remains unresolved. We give a fully explicit primal–dual linear programming formulation, explain exactly what the familiar $\Delta = 2$ branch–leaf identity does and does not imply, and give a faithful reconstruction of the two-path upper-bound argument of Cembrano, Fischer, and Klimm. We then prove that the particular projected system of inequalities used in that argument is sharp: no stronger upper bound can follow from those displayed inequalities alone. Finally, exact rational primal and dual certificates establish

$$(\alpha_3^*, \alpha_4^*, \alpha_5^*, \alpha_6^*, \alpha_7^*) = \left(\frac{5}{6}, \frac{3}{4}, \frac{3}{4}, \frac{35}{48}, \frac{76}{105} \right).$$

In particular, the published upper bound $76/105$ is attainable when $n = 7$. This is a finite-instance result and does not imply asymptotic $3/4$ -optimality.

Outcome. The work below gives verified partial progress. It neither constructs mechanisms whose guarantees converge to $3/4$ nor proves a constant upper bound below $3/4$.

1 The problem and the status of the question

Let $V = [n]$. A graph $G \in \mathcal{G}_n$ is a directed graph on V with no loops and exactly one outgoing edge from every vertex. Equivalently, it is a function $g : V \rightarrow V$ with $g(v) \neq v$. Let $d_G^-(v)$ be the indegree of v and let

$$\Delta(G) = \max_{v \in V} d_G^-(v).$$

A randomized selection mechanism assigns to every $G \in \mathcal{G}_n$ a probability vector $f(G) = (f_v(G))_{v \in V}$. It is *impartial* if changing only the outgoing edge of v cannot change $f_v(G)$.

For fixed n , define the optimal worst-case guarantee

$$\alpha_n^* = \sup_f \min_{G \in \mathcal{G}_n} \frac{\sum_{v \in V} f_v(G) d_G^-(v)}{\Delta(G)},$$

where the supremum is over randomized impartial mechanisms. The supremum is in fact a maximum by the finite LP in theorem 2.1. Cembrano, Fischer, and Klimm prove a uniform lower bound $2105/3147$ and, for every $n \geq 6$, the upper bound

$$U_n := \frac{3n^3 - 19n^2 + 30n - 4}{4n(n-2)(n-4)} = \frac{3}{4} - \frac{n^2 - 6n + 4}{4n(n-2)(n-4)}. \quad (1)$$

Thus $\limsup_n \alpha_n^* \leq 3/4$, while the cited work leaves open whether $\alpha_n^* \rightarrow 3/4$ [1].

The purpose of this note is not to claim a resolution. It replaces several unsupported statements in an earlier draft by two precise results:

- (i) the projected inequality system used in the published two-path proof has an exactly computable optimum, equal to the published bound; and
- (ii) the complete anonymous LP has exact rational primal and dual certificates through $n = 7$.

2 An exact primal–dual formulation

For $v \in V$, let G^{-v} denote the partial graph obtained by deleting the outgoing edge of v and retaining every other edge. Let $\mathcal{H}_{n,v}$ be the set of all such partial graphs with distinguished vertex v .

Proposition 2.1 (Exact primal LP). *For fixed n , α_n^* is the optimum of*

$$\begin{aligned} \max \quad & \alpha \\ \text{subject to} \quad & \sum_{v \in V} x_{v,G^{-v}} = 1 && \text{for every } G \in \mathcal{G}_n, \end{aligned} \tag{2}$$

$$\begin{aligned} & \sum_{v \in V} d_G^-(v) x_{v,G^{-v}} \geq \alpha \Delta(G) && \text{for every } G \in \mathcal{G}_n, \\ & x_{v,H} \geq 0 && \text{for every } v \in V, H \in \mathcal{H}_{n,v}. \end{aligned} \tag{3}$$

The mechanism represented by a feasible point is $f_v(G) = x_{v,G^{-v}}$, and every impartial mechanism has this form.

Proof. The quantity $x_{v,G^{-v}}$ is unchanged when only the outgoing edge of v is changed, so it defines an impartial selection probability. Equation (2) makes these probabilities sum to one and equation (3) is precisely the desired guarantee. Conversely, if f is impartial and two graphs G, G' satisfy $G^{-v} = G'^{-v}$, then $f_v(G) = f_v(G')$. Hence $x_{v,H} := f_v(G)$ is well defined for any completion G of H , and the two families of constraints follow. $\square$

The dual is useful both conceptually and computationally. It also shows why a dual certificate is not merely a probability distribution over bad graphs.

Proposition 2.2 (Exact dual LP). *The dual of the LP in theorem 2.1 is*

$$\begin{aligned} \min \quad & \sum_{G \in \mathcal{G}_n} y_G \\ \text{subject to} \quad & \sum_{G: G^{-v}=H} y_G \geq \sum_{G: G^{-v}=H} z_G d_G^-(v) && \text{for every } v \in V, H \in \mathcal{H}_{n,v}, \end{aligned} \tag{4}$$

$$\begin{aligned} & \sum_{G \in \mathcal{G}_n} z_G \Delta(G) = 1, \\ & z_G \geq 0, \quad y_G \in \mathbb{R} && \text{for every } G \in \mathcal{G}_n. \end{aligned} \tag{5}$$

Strong duality holds.

Proof. For weak duality, let (x, α) be primal feasible and (y, z) dual feasible. By equation (5) and the performance constraints,

$$\begin{aligned} \alpha &= \sum_G z_G \alpha \Delta(G) \leq \sum_G z_G \sum_v d_G^-(v) x_{v, G^{-v}} \\ &= \sum_{v, H} x_{v, H} \sum_{G: G^{-v}=H} z_G d_G^-(v) \\ &\leq \sum_{v, H} x_{v, H} \sum_{G: G^{-v}=H} y_G = \sum_G y_G \sum_v x_{v, G^{-v}} = \sum_G y_G. \end{aligned}$$

The displayed program is the standard LP dual of equations (2) and (3); since the primal is a finite feasible LP with bounded objective, strong duality applies. $\square$

Proposition 2.3 (Symmetrization). *For every impartial mechanism there is an anonymous impartial mechanism with at least the same guarantee. Consequently, α_n^* is unchanged if mechanisms are required to be invariant under relabeling of the vertices.*

Proof. Average the mechanism over all $n!$ relabelings. Impartiality and normalization are preserved. For any fixed graph, the expected selected indegree under the averaged mechanism is the average of the corresponding quantities on isomorphic relabelings, each of which satisfies the original worst-case guarantee. $\square$

3 The $\Delta = 2$ identity is not a reduction

For a graph with $\Delta(G) = 2$, define

$$P_j(G) = \sum_{v: d_G^-(v)=j} f_v(G), \quad j = 0, 1, 2.$$

Proposition 3.1 (Branch-leaf identity). *If $\Delta(G) = 2$, then*

$$\frac{\sum_v f_v(G) d_G^-(v)}{2} = \frac{1 - P_0(G) + P_2(G)}{2}. \quad (6)$$

Hence a mechanism has ratio at least $3/4 - \varepsilon$ on this particular graph if and only if

$$P_2(G) - P_0(G) \geq \frac{1}{2} - 2\varepsilon. \quad (7)$$

Proof. Since $P_0 + P_1 + P_2 = 1$, the expected indegree equals $P_1 + 2P_2 = 1 - P_0 + P_2$. $\square$

Remark 3.2 (Logical limitation). Theorem 3.1 is an exact rewriting of the objective on $\Delta = 2$ graphs. It does not prove that such graphs determine α_n^* , and it gives no performance guarantee on graphs with $\Delta \geq 3$. In particular, checking equation (7) on all $\Delta = 2$ graphs is not, by itself, a reduction of the asymptotic problem.

4 The two-path proof, reconstructed correctly

We reproduce the part of the upper-bound argument needed for the subsequent sharpness statement. Fix $n \geq 6$ and put

$$m = \left\lfloor \frac{n}{2} \right\rfloor - 1.$$

For $i = 0, \dots, m$, let $G_i = (V, E_i)$, where

$$\begin{aligned} E_i = & \{(v+1, v) : v \in \{1, \dots, n-1\} \setminus \{2, i+2\}\} \\ & \cup \{(1, 2), (3, 1 + \mathbf{1}_{\{i=0\}}), (i+3, 2)\}. \end{aligned} \quad (8)$$

These graphs consist of a directed 2-cycle and two directed paths entering its two vertices. We may assume the mechanism is anonymous by theorem 2.3. Put

$$p_v = f_v(G_0), \quad x_i = f_2(G_i) \quad (i = 1, \dots, m).$$

The rooted deleted-edge isomorphisms used in [1] give

$$p_1 = \frac{1}{n}, \quad p_3 \leq \frac{1}{n-2}, \quad (9)$$

$$f_2(G_i) = f_1(G_{i+1}) \quad (i = 0, \dots, m-1), \quad (10)$$

$$f_3(G_i) = p_{n-i+1}, \quad f_{i+3}(G_i) = p_{i+3} \quad (i = 1, \dots, m). \quad (11)$$

For completeness, the first equality in equation (9) follows by comparing $(1, G_0^{-1})$ with a directed n -cycle rooted at one vertex. The second comparison is with a disjoint union of a directed 2-cycle and a directed $(n-2)$ -cycle; anonymity makes the $n-2$ vertices of the latter equiprobable. The relations equations (10) and (11) follow by deleting the indicated outgoing edge: in each comparison the resulting rooted graph is the same pair of directed paths, up to relabeling. This is also immediate from equation (8).

Set

$$B_n := 1 - \frac{1}{n} - \frac{1}{n-2} = \frac{n^2 - 4n + 2}{n(n-2)}. \quad (12)$$

Using nonnegativity and that probabilities sum to one in each G_i , equations (9) to (11) imply

$$p_2 + \sum_{v=4}^n p_v \geq B_n, \quad (13)$$

$$p_2 + p_4 + p_n + x_1 \leq 1, \quad (14)$$

$$p_{n-i+1} + p_{i+3} + x_{i-1} + x_i \leq 1 \quad (i = 2, \dots, m). \quad (15)$$

Notice that the index set is $i = 0, \dots, m$, not $i = 0, \dots, n-4$.

For $i = 1, \dots, m$, form G'_i by changing the outgoing edge of vertex 2 from $(2, 1)$ to $(2, n)$. Vertex 2 is then the *unique* vertex of indegree two, while every other vertex has indegree at most one. Impartiality therefore gives

$$\alpha \leq \frac{1+x_i}{2} \quad (i = 1, \dots, m), \text{ and hence } \alpha \leq \frac{1 + \min_i x_i}{2}. \quad (16)$$

The uniqueness of the maximum-indegree vertex is essential; the endpoint argument is not an argument about two maximum-indegree candidates.

Summing equations (14) and (15) with the parity-dependent weights in the published proof yields equation (1). The next result says exactly how much information those displayed inequalities contain.

Proposition 4.1 (Sharpness of the projected two-path system). *Let $\mathcal{R}_n$ be the set of nonnegative vectors $(p_2, p_4, \dots, p_n, x_1, \dots, x_m)$ satisfying equations (13) to (15), and define*

$$t_n := \max_{(p,x) \in \mathcal{R}_n} \min_{1 \leq i \leq m} x_i.$$

Then

$$t_n = \frac{1}{2(n-4)} (n-3-2B_n) = 2U_n - 1. \quad (17)$$

Consequently, equations (13) to (16) imply exactly $\alpha \leq U_n$, and cannot imply a strictly smaller upper bound.

Proof. First suppose that $n = 2m + 3$ is odd. Each of the variables $p_2, p_4, \dots, p_n$ appears exactly once when equations (14) and (15) are summed. Hence

$$B_n + 2 \sum_{i=1}^{m-1} x_i + x_m \leq m.$$

The total weight on the x_i is $2m - 1 = n - 4$, so

$$\min_i x_i \leq \frac{m - B_n}{2m - 1} = \frac{n - 3 - 2B_n}{2(n - 4)}. \quad (18)$$

Now suppose that $n = 2m + 2$ is even. Multiply equation (14) and equation (15) for $i = 2, \dots, m-1$ by two, leave the final constraint $i = m$ with weight one, and add. Every p -variable then has coefficient two, including the central variable $p_{m+3} = p_{n-m+1}$, which occurs twice in the final constraint. Therefore

$$2B_n + 4 \sum_{i=1}^{m-2} x_i + 3x_{m-1} + x_m \leq 2m - 1.$$

The total weight on the x_i is $4m - 4 = 2(n - 4)$, giving the same upper bound as in equation (18).

It remains to prove attainability in the relaxation. Let t denote the right-hand side of equation (17) and set $x_i = t$ for every i . The definition gives $t \geq 0$, and $B_n \geq 1/2$ for $n \geq 6$ implies $t \leq 1/2$. Hence t , $1 - t$, and $1 - 2t$ are nonnegative. If $n = 2m + 3$ is odd, set

$$p_2 = 1 - t, \quad p_{i+3} = 1 - 2t \quad (i = 2, \dots, m),$$

and set every other p -variable in equation (13) to zero. Then every constraint equations (14) and (15) is tight and

$$p_2 + \sum_{v=4}^n p_v = 1 - t + (m-1)(1-2t) = m - (2m-1)t = B_n.$$

If $n = 2m + 2$ is even, set

$$p_2 = 1 - t, \quad p_{i+3} = 1 - 2t \quad (i = 2, \dots, m-1), \quad p_{m+3} = \frac{1-2t}{2},$$

and set all remaining p -variables to zero. Again every constraint is tight, and

$$p_2 + \sum_{v=4}^n p_v = 1 - t + \left(m - \frac{3}{2}\right)(1-2t) = B_n.$$

Thus $t_n \geq t$, which together with the upper bounds proves equation (17). Finally, substituting t_n into equation (16) gives U_n by direct algebra. $\square$

Remark 4.2 (What has and has not been ruled out). Theorem 4.1 rigorously rules out improving the published bound by manipulating only equations (13) to (15) and the endpoint inequality. It does *not* prove that the entire two-path graph family, with every impartiality relation and every performance constraint included, is incapable of yielding a stronger bound. It also does not constrain dual certificates supported on other graph families.

5 The anonymous orbit LP and exact certificates through $n = 7$

We now make the finite computations exact. Let $\mathcal{O}_n$ be the set of isomorphism classes of graphs in $\mathcal{G}_n$, and let $\mathcal{T}_n$ be the set of isomorphism classes of pairs (v, G^{-v}) with the vertex v distinguished. For a representative $G \in \mathcal{O}_n$ and $\tau \in \mathcal{T}_n$, define

$$M_{G,\tau} = |\{v \in V : (v, G^{-v}) \text{ has type } \tau\}|,$$

$$Q_{G,\tau} = \sum_{\substack{v \in V: \\ (v, G^{-v}) \text{ has type } \tau}} d_G^-(v).$$

By theorem 2.3, the exact anonymous orbit LP is

$$\begin{aligned} & \max \quad \alpha \\ & \text{subject to} \quad \sum_{\tau \in \mathcal{T}_n} M_{G,\tau} x_\tau = 1 \end{aligned} \quad (G \in \mathcal{O}_n), \quad (19)$$

$$\begin{aligned} & \sum_{\tau \in \mathcal{T}_n} Q_{G,\tau} x_\tau \geq \alpha \Delta(G) \end{aligned} \quad (G \in \mathcal{O}_n), \quad (20)$$

$$x_\tau \geq 0 \quad (\tau \in \mathcal{T}_n).$$

Its dual is

$$\begin{aligned} & \min \quad \sum_{G \in \mathcal{O}_n} y_G \\ & \text{subject to} \quad \sum_{G \in \mathcal{O}_n} M_{G,\tau} y_G \geq \sum_{G \in \mathcal{O}_n} Q_{G,\tau} z_G \end{aligned} \quad (\tau \in \mathcal{T}_n), \quad (21)$$

$$\begin{aligned} & \sum_{G \in \mathcal{O}_n} \Delta(G) z_G = 1, \end{aligned} \quad (22)$$

$$z_G \geq 0, \quad y_G \in \mathbb{R} \quad (G \in \mathcal{O}_n).$$

5.1 Canonical enumeration

The orbit enumeration uses the standard decomposition of a functional digraph. Every weak component has one directed cycle; after removing the cycle arcs, an unordered rooted in-tree is attached to every cycle vertex. A rooted tree is encoded recursively by the sorted multiset of its child codes. A component is encoded by the cyclic word of its rooted-tree codes, minimized over cyclic rotations, and a full graph is the sorted multiset of its component codes. In a partial graph G^{-v} , the component containing the distinguished vertex v is a rooted tree, while every other component is encoded as above. This gives complete isomorphism invariants for both $\mathcal{O}_n$ and $\mathcal{T}_n$.

The resulting orbit counts and exact optimum values are as follows.

n	$ \mathcal{O}_n $	$ \mathcal{T}_n $	α_n^*
3	2	3	5/6
4	6	7	3/4
5	13	19	3/4
6	40	47	35/48
7	100	130	76/105

Table 1: Exact anonymous orbit LP values.

Theorem 5.1 (Exact finite values through seven vertices). *The values in table 1 are exact. In particular, there exists an impartial mechanism on seven vertices with guarantee 76/105, and no impartial mechanism on seven vertices has a larger guarantee.*

Proof. The ancillary file `certificates_n3_n7.json` contains, for every $n = 3, \dots, 7$, nonnegative rational values x_τ satisfying equations (19) and (20) at the value in table 1. It also contains rational values (y_G, z_G) satisfying equations (21) and (22), with $z_G \geq 0$ and dual objective equal to the same value. Weak duality therefore proves optimality.

The standard-library script `verify_certificates.py` independently reconstructs all graph and partial-graph orbits from the canonical decomposition above and checks every equality and inequality using exact `Fraction` arithmetic. For integrity, the SHA-256 digest of the JSON certificate supplied with this note is

0c694fc4d6eaa23a6c213a5d342399f2bd296c798d9cb7e4f4a6ee103123c150.

Thus the lower and upper certificates are finite, exact, and reproducible. $\square$

Remark 5.2. The equality $\alpha_7^* = 76/105$ solves the fixed-size optimization problem for $n = 7$. It is not an asymptotic construction: the certificate consists of probabilities for the 130 rooted partial-graph types that occur at that single value of n .

6 What remains open

The asymptotic question requires a mechanism family, or an equivalent family of feasible LP solutions, for unbounded n . None of the results above provides such a family. The exact finite certificates suggest that the published two-path upper bound can be tight at small values of n , but they do not reveal a stable formula for the probabilities $x_{v,H}$ as n grows.

A genuine solution in the positive direction would need to construct impartial mechanisms $f^{(n)}$ satisfying

$$\min_{G \in \mathcal{G}_n} \frac{\sum_v f_v^{(n)}(G) d_G^-(v)}{\Delta(G)} \geq \frac{3}{4} - o(1).$$

A genuine negative solution would need a sequence of valid dual certificates whose objectives are at most $3/4 - c$ for some fixed $c > 0$. By theorem 4.1, such a negative result cannot be obtained from the displayed projected two-path inequalities alone. It would have to use additional impartiality relations, additional performance constraints, or a richer graph family.

Conclusion

The corrected assessment is therefore precise. The exact LP formulation and the $\Delta = 2$ identity are valid, but the identity is not a reduction of the open problem. The published two-path proof uses the parity-dependent weighted inequalities above and an endpoint graph with a unique maximum-indegree vertex. The resulting projected relaxation is sharp. Finally, exact primal–dual certificates determine the optimal values through $n = 7$, including $\alpha_7^* = 76/105$. These are verified partial results; the asymptotic $3/4$ problem remains unsolved here.

References

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